C.2 Joint Moments

L1 Means of Multi-Variate Distributions

- For discrete RVs,

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) \cdot P(X=x, Y=y)$$

if,

$$E[X] = \sum_x \sum_y x \cdot P(X=x, Y=y)$$

$$E[X^2] = \sum_x \sum_y x^2 \cdot P(X=x, Y=y)$$

Eg. $p(x,y) = \frac{2x+y}{12}$

for $(x,y) = (0,1), (0,2), (1,2), (1,3)$

$E[X] = ?$

$$E[X] = 0 \cdot P(X=0, Y=1) + 0 \cdot P(X=0, Y=2) + 1 \cdot P(X=1, Y=2) + 1 \cdot P(X=1, Y=3)$$

$$= \frac{9}{12} = \frac{3}{4}$$

L2 Covariances & Correlations

$$Var(X) = E[X^2] - (E[X])^2$$

$$\boxed{Cov(X,Y) = E[X \cdot Y] - E[X] \cdot E[Y]}$$

$$Cov(X, b) = E[bX] - b\,E[X] = 0$$

- Covariance is bilinear -

$$Cov(aX + bY, cZ) = Cov(aX, cZ) + Cov(bY, cZ)$$
$$= ac\,Cov(X, Z) + bc\,Cov(Y, Z)$$

- So impt:

$$\boxed{Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)}$$

$$Var(X-Y) = Var(X) + Var(Y) - 2\,Cov(X,Y)$$

If X & Y are independent.

$E[X \cdot Y] = E[X] \cdot E[Y]$, then $cov = 0$.

$$Var(X+Y) = Var(X) + Var(Y)$$
$$Var(X-Y) = Var(X) + Var(Y)$$

- The correlation of X and Y,

$$\boxed{Corr(X,Y) = \frac{Cov(X,Y)}{SD(X) \cdot SD(Y)}}$$

If X & Y are independent, the $cov(X,Y) = 0$
then $corr(X,Y) = 0$

- Properties of correla^n.

→ $Y = aX + b$

$$\begin{aligned} Cov(X, Y) &= Cov(X, aX + b) \\ &= a\, Cov(X, X) + \underbrace{Cov(X, b)}_{0} \\ &= a\, Var(X) \end{aligned}$$

→ $-1 \leq Corr(X, Y) \leq 1$.

- X & Y are independent if both
1. Support of (X, Y), points s.t. $P(X = x, Y = y) > 0$, is a rectangular lattice.
2. $P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$

If X & Y are independent, then,

$$E[XY] = E[X] \cdot E[Y]$$
$$Cov(X, Y) = 0$$

Remember, Independence $\Rightarrow$ $Cov(X, Y) = 0$

the converse is not always true.

L3 Conditional Moments

- The conditional expecta^n of X is a func^n of Y.

$$E[X \mid Y = y] = \sum_{\text{all } x} x \cdot P(X = x \mid Y = y)$$

$$\boxed{E[X] = E[E[X \mid Y = y]]}$$

$$= \sum_{\text{all } y} E[X \mid Y = y] \cdot P(Y = y)$$

↳ This is particularly helpful when distribuⁿ of X is complicated, but the conditional distribuⁿ $(X \mid Y)$ & distribuⁿ of Y are both nice.

- $E[X] = E[E[X \mid Y]]$

$E[X^2] = E[E[X^2 \mid Y]]$

but this is not true for variation.

- Law of Total Variaⁿ

$$Var(X) = E[Var(X \mid Y)] + Var[E(X \mid Y)]$$

If $S = X_1 + \cdots + X_n$

$E[S \mid N] = N\, E[X]$

$Var(S \mid N) = N \cdot Var(X)$

$$\boxed{Var(S) = E[N]\, Var(X) + Var(N)\, (E(X))^2}$$